Finding the Inverse of a Function Algebraically

Algebra 2 Worksheet · Grade 9–11

Name: _____ Date: _____

Learning Objectives

- Find the inverse of a function by swapping x and y and solving for y
- Apply domain restrictions when determining the correct form of an inverse function
- Verify inverse functions using composition

Problems

1. Find the inverse of the function below:

$f(x) = x + 5$

2. Find the inverse of the function below:

$f(x) = 3x - 7$

3. Find the inverse of the function below, given that the domain is all x greater than or equal to 0:

$f(x) = x^2 + 3$

4. Find the inverse of the function below:

$f(x) = x^3 + 2$

5. Find the inverse of the function below:

$f(x) = \sqrt{x - 4}$

6. Find the inverse of the function below:

$f(x) = (x + 2)^3$

7. Find the inverse of the function below, given that x is less than or equal to 2:

$$f(x) = (x-2)^2 + 1$$

8. Find the inverse of the function below:

$$f(x) = \frac{2x+1}{3}$$

9. Find the inverse of the function below:

$$f(x) = \sqrt[3]{x+5} - 2$$

10. Find the inverse of the function below, then verify by computing f composed with f-inverse:

$$f(x) = \frac{x-1}{x+2}$$

Finding the Inverse of a Function Algebraically — Answer Key

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Answer Key

1. Answer: $f^{-1}(x) = x - 5$

- Replace f(x) with y: $y = x + 5$
- Swap x and y: $x = y + 5$
- Solve for y: $y = x - 5$
- Write as inverse: $f^{-1}(x) = x - 5$

2. Answer: $f^{-1}(x) = (x + 7) / 3$

- Replace f(x) with y: $y = 3x - 7$
- Swap x and y: $x = 3y - 7$
- Add 7 to both sides: $x + 7 = 3y$
- Divide both sides by 3: $y = (x + 7) / 3$
- Write as inverse: $f^{-1}(x) = (x + 7) / 3$

3. Answer: $f^{-1}(x) = \sqrt{(x - 3)}$

- Replace f(x) with y: $y = x^2 + 3$
- Swap x and y: $x = y^2 + 3$
- Subtract 3 from both sides: $x - 3 = y^2$
- Take the square root of both sides: $y = \pm\sqrt{(x - 3)}$
- Since domain restriction $x \geq 0$, take positive root: $f^{-1}(x) = \sqrt{(x - 3)}$

4. Answer: $f^{-1}(x) = \sqrt[3]{(x - 2)}$

- Replace f(x) with y: $y = x^3 + 2$
- Swap x and y: $x = y^3 + 2$
- Subtract 2 from both sides: $x - 2 = y^3$
- Take the cube root of both sides: $y = \sqrt[3]{(x - 2)}$
- Write as inverse: $f^{-1}(x) = \sqrt[3]{(x - 2)}$

5. Answer: $f^{-1}(x) = x^2 + 4$

- Replace f(x) with y: $y = \sqrt{(x - 4)}$
- Swap x and y: $x = \sqrt{(y - 4)}$
- Square both sides: $x^2 = y - 4$
- Add 4 to both sides: $y = x^2 + 4$
- Write as inverse: $f^{-1}(x) = x^2 + 4$

6. Answer: $f^{-1}(x) = \sqrt[3]{x} - 3$

- Replace f(x) with y: $y = (x + 3)^3$
- Swap x and y: $x = (y + 3)^3$

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- Take the cube root of both sides: $\sqrt[3]{x} = y + 3$
- Subtract 3 from both sides: $y = \sqrt[3]{x} - 3$
- Write as inverse: $f^{-1}(x) = \sqrt[3]{x} - 3$

7. Answer: $f^{-1}(x) = 2 - \sqrt{(x - 1)}$

- Replace f(x) with y: $y = (x - 2)^2 + 1$
- Swap x and y: $x = (y - 2)^2 + 1$
- Subtract 1: $x - 1 = (y - 2)^2$
- Take the square root: $\pm\sqrt{(x - 1)} = y - 2$
- Since domain restriction $x \leq 2$, use negative root: $y = 2 - \sqrt{(x - 1)}$
- Write as inverse: $f^{-1}(x) = 2 - \sqrt{(x - 1)}$

8. Answer: $f^{-1}(x) = (3x - 1) / 2$

- Replace f(x) with y: $y = (2x + 1) / 3$
- Swap x and y: $x = (2y + 1) / 3$
- Multiply both sides by 3: $3x = 2y + 1$
- Subtract 1: $3x - 1 = 2y$
- Divide by 2: $y = (3x - 1) / 2$
- Write as inverse: $f^{-1}(x) = (3x - 1) / 2$

9. Answer: $f^{-1}(x) = (x + 2)^3 - 5$

- Replace f(x) with y: $y = \sqrt[3]{(x + 5)} - 2$
- Swap x and y: $x = \sqrt[3]{(y + 5)} - 2$
- Add 2 to both sides: $x + 2 = \sqrt[3]{(y + 5)}$
- Cube both sides: $(x + 2)^3 = y + 5$
- Subtract 5: $y = (x + 2)^3 - 5$
- Write as inverse: $f^{-1}(x) = (x + 2)^3 - 5$

10. Answer: $f^{-1}(x) = (-2x - 1) / (x - 1)$

- Replace f(x) with y: $y = (x - 1) / (x + 2)$
- Swap x and y: $x = (y - 1) / (y + 2)$
- Multiply both sides by (y + 2): $x(y + 2) = y - 1$
- Expand: $xy + 2x = y - 1$
- Collect y terms: $xy - y = -2x - 1$
- Factor: $y(x - 1) = -2x - 1$
- Divide: $y = (-2x - 1) / (x - 1)$
- Write as inverse: $f^{-1}(x) = (-2x - 1) / (x - 1)$
- Verify: $f(f^{-1}(x)) = x$ ✓